INCH-POUND

MIL-C-18757/5(OS)
19 June 1991
SUPERSEDING
WS 16575D
13 May 1988

MILITARY SPECIFICATION

CARTRIDGE CASE, STEEL, 76-MILLIMETER (62-CALIBER)

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the 76-millimeter, 62-caliber steel cartridge case, herein referred to as the "cartridge case."

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-C-18757	Cartridge Cases, Steel (Other Than Small Arms), General Specification for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Branch (Code 3730), Indian Head, MD 20640-5000 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or sending a letter.

AMSC N/A FSC 1395

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STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage
DOD-STD-2101	Classification of Characteristics

(Unless otherwise indicated, copies of federal and military specifications and standards are available from: Standardization Documents Order Desk , Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

NAVAL SEA SYSTEMS COMMAND (CAGE Codes 10001 and 53711)

3028547	Case, Cartridge, 76mm/62 Caliber, Steel
3030497	Receiver, 76mm/62 Caliber Case Body Gage
5366338	Cartridge, 76mm/62 Caliber, Mk 201 Mod 1

(Application for copies of NAVSEA drawings should be addressed to: Commanding Officer, Naval Ordnance Station (Attn: Code 802), Louisville, KY 40214-5001.)

PUBLICATIONS

NAVAL SEA SYSTEMS COMMAND (CAGE Code 10001)

OD 47471	Method for Rating the Effects of Salt Spray Testing on Zinc Electroplated Cartridge Cases

(Application for copies should be addressed to: Commanding Officer, Naval Ship Weapon Systems Engineering Station, NAVSEA Data Support Activity (Code 5H30), Port Hueneme, CA 93043.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E8 Tension Testing of Metallic Materials, Standard Test Methods of (DOD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187).

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 **Order of precedence.** In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 **First article sample.** When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.4.

3.2 **Characteristics.** Unless otherwise specified in the contract or purchase order, the cartridge case shall conform to the requirements specified herein and on Drawing 3028547. The general requirements of MIL-C-18757 shall also be applicable. Where the requirements of MIL-C-18757 and this specification conflict, the requirements of this specification shall take precedence (see 4.6.1).

3.3 **High stress expansion (M101).** The cartridge case shall show no evidence of cracks in any portion of the case when subjected to the high stress expansion test of 4.6.2.

3.4 **Hardness (M102).** The cartridge cases shall conform to the hardness values specified in table I in the regions shown in figure 1. No individual readings shall be outside the regional requirement (see 4.6.3).

TABLE I. Hardness and tensile strength.

Region	Hardness[1]		Yield Strength at 2% offset, (psi)	Elongation (1" gage length)
	Max	Min		
B	C42	C32	135,000 Min	5% Min
C	C41	B96	90,000 Min	6% Min
D	C30	B94	———	———
E	B85	—	———	———

[1] Rockwell "C" or "B" scale.

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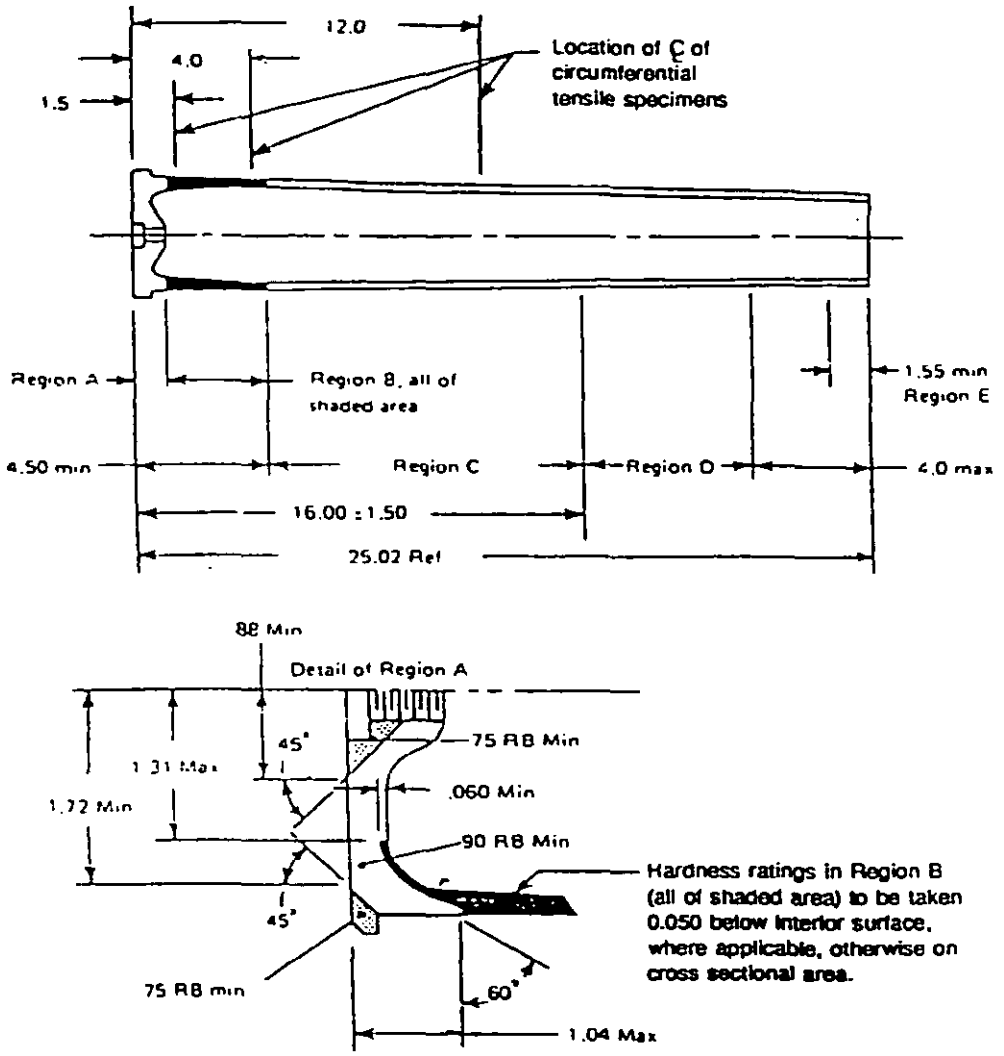


FIGURE 1. Hardness/tensile specimen location.

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3.5 Tensile strength and elongation (M103). The yield strength and elongation values for the cartridge case shall be as specified in table I in the regions shown in figure 1 (see 4.6.4 and 4.6.5).

3.6 Protective coating (M104). The cartridge case protective coating shall have a rating of the black corrosion products of zinc less than the established standard of 0.24000 (see 4.6.6).

3.7 Chamber gage (M105). Each cartridge case shall fully chamber in a Navy approved chamber gage (see 4.6.7).

3.8 Ballistics (M106). Each cartridge case shall load, eject, show no splits, and otherwise perform satisfactorily when ballistically tested (see 4.6.8).

3.9 Workmanship. Workmanship shall be as specified in MIL-C-18757 (see 4.6.1).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 *Responsibility for compliance.* *All items shall meet all requirements of sections 3 and 5.* The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)
b. Quality conformance inspection (see 4.5)

4.3 Classification of characteristics. The characteristics verified by the tests and examinations specified herein and in the detail inspection are classified as critical, major or minor in accordance with DOD-STD-2101. Requirements for critical characteristics are identified by the symbol (C), and major characteristics by the symbol (M). The number following the classification symbol indicates the serial number of the characteristic. Requirements which are not annotated with a classification code are classified as minor.

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4.4 First article inspection. The first article sample shall consist of 158 cartridge cases and shall be examined and tested in accordance with table I of MIL-C-18757. The cartridge cases shall be the same as for regular production and manufactured under the same conditions as the production lot. Further production of cartridge cases by the contractor, prior to the approval of the first article, shall be at the contractor's risk.

4.4.1 First article sample acceptance/rejection. Acceptance or rejection of the first article sample shall be in accordance with table I of MIL-C-18757. For those where the acceptance criteria is not shown, any failure to meet the requirements shall be cause for rejection of the sample.

4.5 Quality conformance inspection. Quality conformance inspection shall consist of all the examinations and tests as specified in table I of MIL-C-18757.

4.5.1 Lot size. Unless otherwise specified in the contract or purchase order (see 6.2), the production lot shall consist of 5000 cartridge cases plus those required for test purposes.

4.6 Methods of inspection.

4.6.1 Visual and dimensional examination. The charges shall be visually and dimensionally examined for conformance to requirements of 3.2 and 3.9. Four power (4X) magnification may be used for examination of defects.

4.6.2 High stress expansion test. Each cartridge case in the lot (see 4.5.1), after heat treatment but prior to tapering, shall be subjected to a high stress expansion test. The cartridge case shall be expanded diametrically 0.050 inch minimum while under pressure, in a zone extending from 1.5 inches above the flange to 9 inches above the flange. Expansion will decrease as the flange is approached with expansion at the flange to be zero.

4.6.3 Hardness. Hardness testing shall be as specified in MIL-C-18757. Hardness readings shall be taken on each of two sections cut longitudinally from diametrically opposite quadrants of each cartridge case.

4.6.4 Elongation test. Circumferential specimens cut from three positions shown in figure 1 shall be tested for elongation in accordance with MIL-C-18757.

4.6.5 Tension test. Circumferential specimens cut from three positions shown in figure 1 shall be tested for yield strength in accordance with ASTM E8. An offset value of 0.002 inch per inch length shall be used to determine yield strength.

4.6.6 Protective coating tests. Cartridge cases shall be prepared and tested in accordance with the protective coating test of MIL-C-18757. The presence of white corrosion products will be permissible to the extent that it does not interfere with chambering of the cartridge case. The black speck corrosion rating procedure shall be as specified in OD 47471.

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4.6.7 Chamber gage examination. Each cartridge case shall be chamber-gaged in receiver, Drawing 3030497, to determine conformance to the requirements of 3.7. A gage removal force of 35 pounds maximum is permitted.

4.6.8 Ballistic test. The cartridge cases shall be loaded in accordance with Drawing 5366338. The propellant charge shall be adjusted so that the ballistic test pressure at any temperature is as follows:

Service rounds 17.0 to 19.0 tsi-copper
Proof rounds 20.8 to 21.8 tsi-copper

Testing shall be performed in a Mk 75 gun mount and in quantities specified in table II.

TABLE II. Ballistic acceptance.

Type of lot	Number of round	Type of round	Round temperature °C
First article	70	Service	32
	30	Proof	32
	25	Service	49
	25	Service	-7
Production	40[1]	Service	32
	10[1]	Proof	32
Additional test	80	Service	32
	20	Proof	32

[1] Test an additional 100 cases if one case fails to eject or extract; reject if two or more defects are found in the total of 150 cases. After firing, cases shall be rechambered in chamber gage.

4.6.8.1 Test procedure. All proof pressure rounds and five service rounds at each temperature, all containing copper crusher gages, shall be fired in the single shot mode to determine compliance with the pressure requirements. The remainder of the cartridges shall be fired in the automatic mode. Temperature of the rounds shall be as specified in table II. Rounds shall be temperature conditioned for a minimum of 24 hours.

4.7 Inspection of packaging. Packing and marking shall be visually examined for conformance to section 5.

5. PACKAGING

5.1 Packing. Packing shall be in accordance with MIL-C-18757 with four cases packed in each box.

5.2 Palletizing. Cases packed in boxes shall be palletized in accordance with standard commercial practice.

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5.3 Marking. In addition to any special marking required by the contract or purchase order (see 6.2), all marking shall be in accordance with MIL-STD-129. An arrow shall be stenciled on each container as specified in MIL-STD-129.

6. NOTES

6.1 Intended use. The 76-mm, 62-caliber steel cartridge cases, when assembled with their appropriate projectiles, primers, and propellant charges form complete rounds which are used as service ammunition in the Mk 75 gun mount.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1 and 2.2)

c. Whether a first article is required (see 3.1)

d. Agency responsible for inspection if different from 4.1

e. Production lot size, if other than as specified in 4.5.1

f. Measurement and testing of coating if different from 4.6.6

g. Special markings required (see 5.3)

h. That the safety precaution requirements of the "Contractor's Safety Manual for Ammunition, Explosives and Related Dangerous Material," DOD 4145.26M, are applicable (see 6.4).

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, a standard production item from the contractor's current inventory (see 3.1) and the number of items to be tested as specified in 4.4. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

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6.4 Safety precautions. The safety precaution requirements of the "Contractors' Safety Manual for Ammunition, Explosives, and Related Dangerous Material" (DOD 4145.26M) are applicable and should be specified in the contract or order as required by the Federal Acquisition Regulations (FAR) 23.3.

NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of "Ammunition and Explosives Ashore" (OP 5) should be made applicable.

6.5 Subject term (key word) listing.

Ballistic cases
Chamber gage

Preparing Activity:
Navy - OS
(Project 1395-N263)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-C-18757/5(OS)	910619

3. DOCUMENT TITLE

CARTRIDGE CASE, STEEL, 76-MILLIMETER (62-CALIBER)

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)
	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)
Commanding Officer Naval Ordnance Station	(1) Commercial: (301)743-4358 (2) AUTOVON: 364-4358
c. ADDRESS (Include Zip Code) Code 3730 Indian Head, MD 20640-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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